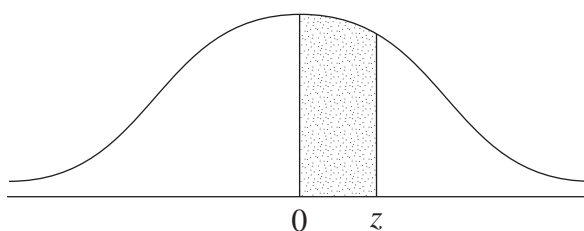


Question 38 (4 marks)

A random variable is normally distributed with mean 0 and standard deviation 1. The table gives the probability that this random variable lies between 0 and z for different values of z .

z	0.1	0.2	0.3	0.4	0.5	0.6
Probability	0.0398	0.0793	0.1179	0.1554	0.1915	0.2257

The probability values given in the table for different values of z are represented by the shaded area in the following diagram.



- (a) Using the table, show that the probability that a value from a random variable that is normally distributed with mean 0 and standard deviation 1 is greater than 0.3 is equal to 0.3821. 1

.....

.....

.....

.....

- (b) Birth weights are normally distributed with a mean of 3300 grams and a standard deviation of 570 grams. By first calculating a z -score, find how many babies, out of 1000 born, are expected to have a birth weight greater than 3471 grams. 3

.....

.....

.....

.....

.....

.....

.....